

1. A station in a network forwards incoming packets by placing them on its shortest output queue. What routing algorithm is being used?
 - A. Hot potato routing
 - B. Static routing
 - C. Flooding
 - D. Delta routing
2. The probability that a single bit will be in error on a typical public telephone line using 4800 bps modem is 10^{-3} . If no error detection mechanism is used, the residual error rate for a communication line using 9-bit frames is approximately equal to
 - A. 0.003
 - B. 0.991
 - C. 0.009
 - D. 0.999
3. Frames from one LAN can be transmitted to another LAN via the device
 - A. Router
 - B. Repeater
 - C. Bridge
 - D. Modem
4. Which of the following is not a disadvantage of wireless LAN?
 - A. lower data transmission
 - B. Higher error rate
 - C. Interference of transmissions from different computers
 - D. All of the above
5. Rs. 449/442-A/423-A is
 - A. A set of physical layer standards developed by EIA and intended to replace Rs- 232-C
 - B. A check bit appended to an array of binary digits to make the sum of the all the binary digits
 - C. A code in which each expression will not conform to specific rules of construction, so that if certain errors occur in an expression the resulting expression will not conform to the rules of construction and thus the presence of the error is detected
 - D. The ratio of the number of data units in error to the total number of data units
6. End-to-end connectivity is provided from host-to-host in:
 - A. Network layer
 - B. Data link layer
 - C. Session layer
 - D. Transport layer
7. A relational operator
 - A. Assigns operand to another
 - B. Compares two operands
 - C. Yields a Boolean result
 - D. Both (B) and (C)
8. A pointer is
 - A. The address of a variable
 - B. The indication of the variable to be accessed next
 - C. A variable for storing address
 - D. Logically combines two operands
9. A friend function can be used to
 - A. Avoid arguments between classes
 - B. Allow access to classes whose source code is unavailable
 - C. Allow one class to access an unrelated class
 - D. Increase the versatility of an overloaded operator

10. In object-oriented programming, advantages of inheritance include
 - A. Avoiding rewriting of code
 - B. Facilitating class libraries
 - C. Providing a useful conceptual framework
 - D. All of the above
11. A constructor is called whenever
 - A. An object is declared
 - B. A class is declared
 - C. An object is used
 - D. A class is used
12. Choose the correct statements
 - A. A destructor is not inherited
 - B. A constructor cannot be called explicitly
 - C. A constructor cannot be virtual
 - D. All of the above
13. CSG can be recognized by a
 - A. FSM
 - B. NDPDM
 - C. DPDM
 - D. Linearly bounded memory machine
14. FSM can recognize
 - A. Any grammar
 - B. Any unambiguous
 - C. Only CG
 - D. Only regular grammar
15. The LR parsing method which is very expensive to implement is
 - A. Simple LR parsing
 - B. Lock ahead LR parsing
 - C. Canonical LR parsing
 - D. None of the above
16. Representation of data structure in memory is known as:
 - A. Abstract data type
 - B. Storage structure
 - C. Recursive
 - D. File structure
17. If the address of A [1][1] and A [2][1] are 1000 and 1010 respectively and each element occupies 2 bytes then the array has been stored in _____.
 - A. Column major
 - B. Matrix
 - C. Row major
 - D. None of the above
18. Quick sort is known as
 - A. Heap sort
 - B. Partition exchange sort
 - C. Merge sort
 - D. Divider sort
19. An algorithm is made up of two independent time complexities $f(n)$ and $g(n)$. then the complexities the algorithm is in the order of _____.
 - A. $f(n) \times g(n)$
 - B. $\min(f(n), g(n))$
 - C. $\max(f(n), g(n))$
 - D. $f(n) + g(n)$
20. Which data structure is needed to convert infix notation to postfix notation?
 - A. Graph
 - B. Queue
 - C. Tree
 - D. Stack
21. The extra key inserted at the end of the array is called a, _____.
 - A. Sentinel
 - B. Stop key
 - C. End key
 - D. Transposition
22. Efficient use of addressing modes
 - A. Speeds up execution
 - B. Reduces the size of instructions
 - C. Reduces the number of instructions
 - D. All of the above

23. Which technique is preferable for transferring a large amount of data to and from a memory in a short time?
 A. DMA
 B. Programmed I/O
 C. Interrupt-driven I/O
 D. None of the above
24. The addressing mode used in an instruction of the form ADD X, Y is
 A. Absolute B. Indirect
 C. Immediate D. Index
25. The shifting of program from one location to another in the memory is called
 A. Relocation
 B. Data transmission
 C. Data collection
 D. DMA
26. Control Memory Address Register is present in
 A. ALU
 B. Control unit
 C. Instruction register unit
 D. Disk control Interface unit
27. The register which contains the data to be written into or read out of the addressed location is known as
 A. Index register
 B. Memory data register
 C. Memory address register
 D. Program counter
28. What is the decimal value of the hexadecimal number 777?
 A. 191 B. 19
 C. 1911 D. 19111
29. How many data select lines are required for selecting eight inputs?
 A. 1 B. 3
 C. 2 D. 4
30. A BCD counter is a
 A. Binary counter
 B. Decade counter
 C. Full-modulus counter
 D. Divide-by-10 counter
31. The most commonly used system for representing signed binary numbers is the:
 A. 2's-complement system
 B. 10's-complement system
 C. 1's-complement system
 D. Sign-magnitude system
32. Select the statement that best describes Read-Only Memory (ROM)
 A. Nonvolatile, used to store information that changes during system operation
 B. Nonvolatile, used to store information that does not change during system operation
 C. Volatile, used to store information that changes during system operation
 D. Volatile, used to store information that does not change during system operation
33. Design phase will usually be
 A. Top-down B. Random
 C. Bottom-up D. All of the above
34. Structure programming codes include
 A. Sequencing B. Iteration
 C. Alteration D. All of the above
35. Which model following testing method is normally used as the acceptance test for a software system?
 A. Regression testing
 B. Unit testing
 C. Integration testing
 D. Function testing

36. Which model reduces the cost of development of software?
 A. Waterfall model
 B. Iterative
 C. Prototyping
 D. None of the above
37. The testing that focuses on the variables is called
 A. Black box testing
 B. Data variable testing
 C. White box testing
 D. Data flow testing
38. Modifying the software to match changes in the ever-changing environment is called
 A. Adaptive maintenance
 B. Perfective maintenance
 C. Corrective maintenance
 D. Preventive maintenance
39. Requirements can be refined using
 A. The water falls model
 B. The evolutionary model
 C. Prototyping model
 D. The spiral model
40. Which of the following command is used to give permission of operation to other users?
 A. Permission B. Grant
 C. Allow D. Revoke
41. Which of the following is the result of a SELECT statement?
 A. TRIGGER B. TABLE
 C. INDEX D. None of these
42. The rule that a value of a foreign key must appear as a value of some specific table is called
 A. Referential constraint
 B. Integrity constraint
 C. Index
 D. Functional dependency
43. _____ refers to a formal process for determining which field belongs in which tables in a relational database.
 A. Normalization
 B. Aggregation
 C. De normalization
 D. None of the above
44. The physical location of a record is determined by a mathematical formula that transforms a file key into record location in
 A. A tree file
 B. A hashed file
 C. An indexed file
 D. A sequential file
45. Which of the following is the recovery management technique in DDBMS?
 A. 2PC (two Phase Commit)
 B. Immediate update
 C. Backup
 D. All of the above
46. Operating system is
 A. A collection of hardware
 B. A collection of software routines
 C. A collection of input-output devices
 D. All of the above
47. The operating system manages
 A. Memory
 B. Disk and I/O devices
 C. Processor
 D. All of the above
48. Round robin is a
 A. Kind of magnetic
 B. Process synchronization policy
 C. Process scheduling policy
 D. Memory allocation policy

49. A set of techniques that allow to executive a program which is not entirely in memory is called
- Demand paging
 - Auxiliary memory
 - Virtual memory
 - Secondary memory
50. Page stealing
- Is a sign of efficient system
 - Should be the tuning goal
 - Is taking page frames other working sets
 - Is taking larger disk spaces for pages paged out
51. Main function of distributed system is
- Resource sharing
 - Reliability and communication
 - Computation speed up
 - All of the above
52. Which of the following statements in true?
- If a language is context free it can always be accepted by a deterministic push-down automaton
 - The union of two context free languages is context free
 - The intersection of two context free language is context free
 - The complement of a context free language is context free
53. Turing Machine (TM) is more powerful than FMS (Finite State Machine) because
- Tape movement is confined to one direction
 - It has no finite state
 - It has the capability to remember arbitrarily long sequences of input symbols
 - None of the above
54. A language L is accepted by a FSA if it is
- CFL
 - Recursive
 - CSL
 - Regular
55. The concept of grammar is much used in this part of the compiler
- Lexical analysis
 - Code generation
 - Parser
 - Code optimization
56. Which of the following is used for grouping of characters into tokens?
- Parser
 - Code Generation
 - Code Optimization
 - Lexical Analyzer
57. A compiler that runs on one machine and produces code for a different machine is called
- Cross Compilation
 - Two pass compilation
 - One pass compilation
 - None of the above
58. Synthesized attribute can be one easily simulated by a
- LL grammar
 - LR grammar
 - Ambiguous grammar
 - None of the above
59. The output of a lexical analyzer is
- Machine code
 - A stream of tokens
 - Intermediate code
 - A parse tree
60. A grammar that produces more than one parse tree for some sentence is called
- Ambiguous
 - Regular
 - Unambiguous
 - None of above

61. Expansion of DDA is _____.
 A. Device Display Analyzer
 B. Digital Device Analyzer
 C. Digital Differential Analyzer
 D. Digital Display Analyzer
62. In beam penetration method, _____ and _____ layers of phosphor are usually used.
 A. Red and green
 B. Yellow and green
 C. Blue and green
 D. Orange and green
63. The scaling transformation alters the size of an _____.
 A. Vector B. Side
 C. Edge D. Object
64. Kirchhoff's second law is based on law of conservation of
 A. Charge B. Momentum
 C. Energy D. Mass
65. Resistivity of a wire depends on
 A. Length
 B. Cross section area
 C. Material
 D. None of the above
66. If two currents are in the same direction at any instant of time in a given branch of a circuit, the net current at that instant
 A. Is zero
 B. Is the sum of the two currents
 C. Is the difference between the two currents
 D. Cannot be determined
67. In "n" type material, majority carriers would be:
 A. Holes B. Slower
 C. Dopants D. Electrons
68. What does a high resistance reading in both forward-and reverse-bias directions indicate?
 A. A good diode
 B. A shorted diode
 C. An open diode
 D. A defective ohmmeter
69. When transistors are used in digital circuits, they usually operate in the
 A. Active region
 B. Saturation and cutoff regions
 C. Breakdown region
 D. Linear region
70. The main feature of a large-signal amplifier is the circuit's _____.
 A. Power efficiency
 B. Maximum power limitations
 C. Impedance matching to the output device
 D. All of the above
71. An example of an analog communication method is
 A. Laser beam
 B. Voice grade telephone line
 C. Micro wave
 D. All of the above
72. Which data communication method is used to send data over a serial communication link?
 A. Simplex B. Full duplex
 C. Half duplex D. All of the above
73. Modulation is the
 A. Varying of some parameter of a carries, such as its amplitude to transmit information
 B. Utilization of a single transmission channel to carrying multiple signals
 C. Transmission of pulses in DC form over a copper wire
 D. None of the above

74. Communications protocols always have a
- Set of symbols
 - Special flag symbol
 - Start of header
 - BCC
75. Which of the following performs modulations and demodulation?
- Fiber optic
 - Coaxial Cable
 - Satellite
 - Modem
76. Mechanism to protect private networks from outside attack is
- Fire wall
 - Digital signature
 - Antivirus
 - Formatting
77. Secret key encryption is also known as
- Asymmetric encryption
 - Secret-encryption
 - Symmetric encryption
 - Private-encryption
78. A computer system that permits multiple users to run programs at same time
- Real time system
 - Time sharing system
 - Multi programming system
 - Multi-tasking system
79. Cryptographic hash function taken an arbitrary block of data and returns
- Fixed size bit string
 - Fixed and variable size bit string
 - Variable size bit string
 - All of the above
80. In asymmetric key cryptography, the private key is kept by:
- Sender
 - Send and receive
 - Receiver
 - All of the above

ANSWER SHEET

1.A	2.C	3.C	4.D	5.A	6.D	7.D	8.C	9.C	10.D
11.A	12.D	13.D	14.D	15.C	16.A	17.C	18.B	19.C	20.D
21.A	22.D	23.A	24.A	25.A	26.B	27.B	28.C	29.B	30.B
31.A	32.B	33.A	34.D	35.D	36.C	37.A	38.A	39.C	40.B
41.B	42.A	43.A	44.B	45.D	46.D	47.D	48.C	49.C	50.C
51.D	52.B	53.C	54.D	55.C	56.D	57.A	58.B	59.B	60.A
61.C	62.A	63.D	64.C	65.C	66.B	67.D	68.C	69.B	70.D
71.D	72.B	73.A	74.A	75.D	76.A	77.C	78.D	79.A	80.C